I_Phi Landscape Analysis — Hubbard L=4 Ne=4 t=1.0 U=4.0 mode=uniform

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System
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  L=4  Ne=4  t=1.0  U=4.0  mode=uniform
  n = N = 4  (pure-state RDM, Tr(rho^2)=1)

  E0  = -2.1027484835
  E1  = -1.8064238518
  Gap = 0.29632463

GME Results
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  C_GME (exact eq.1)   = 0.91287093
  C_LB  (basis states) = 0.66666667
  Gap   = C_GME - C_LB = 0.24620426

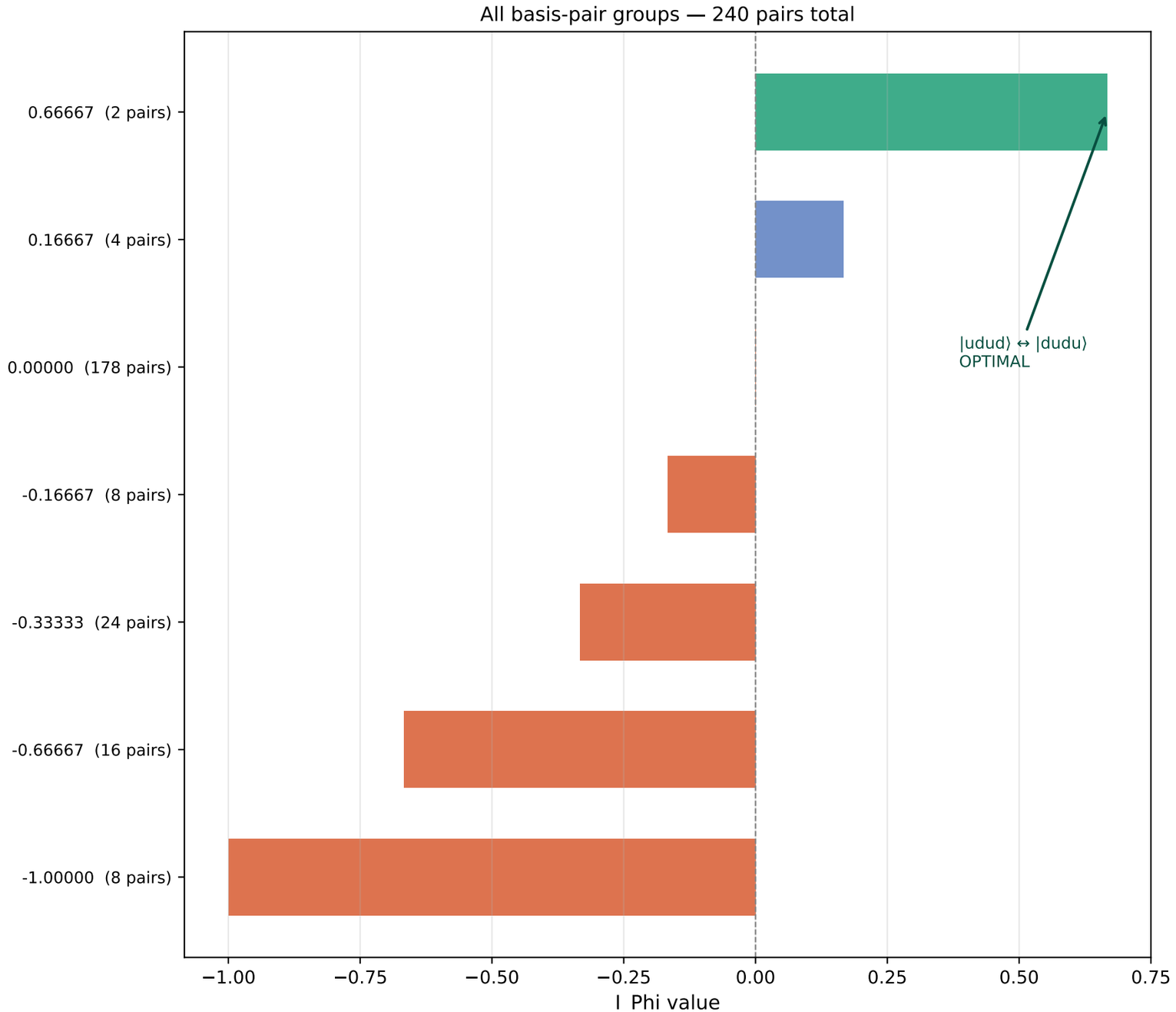
Optimal |Phi>
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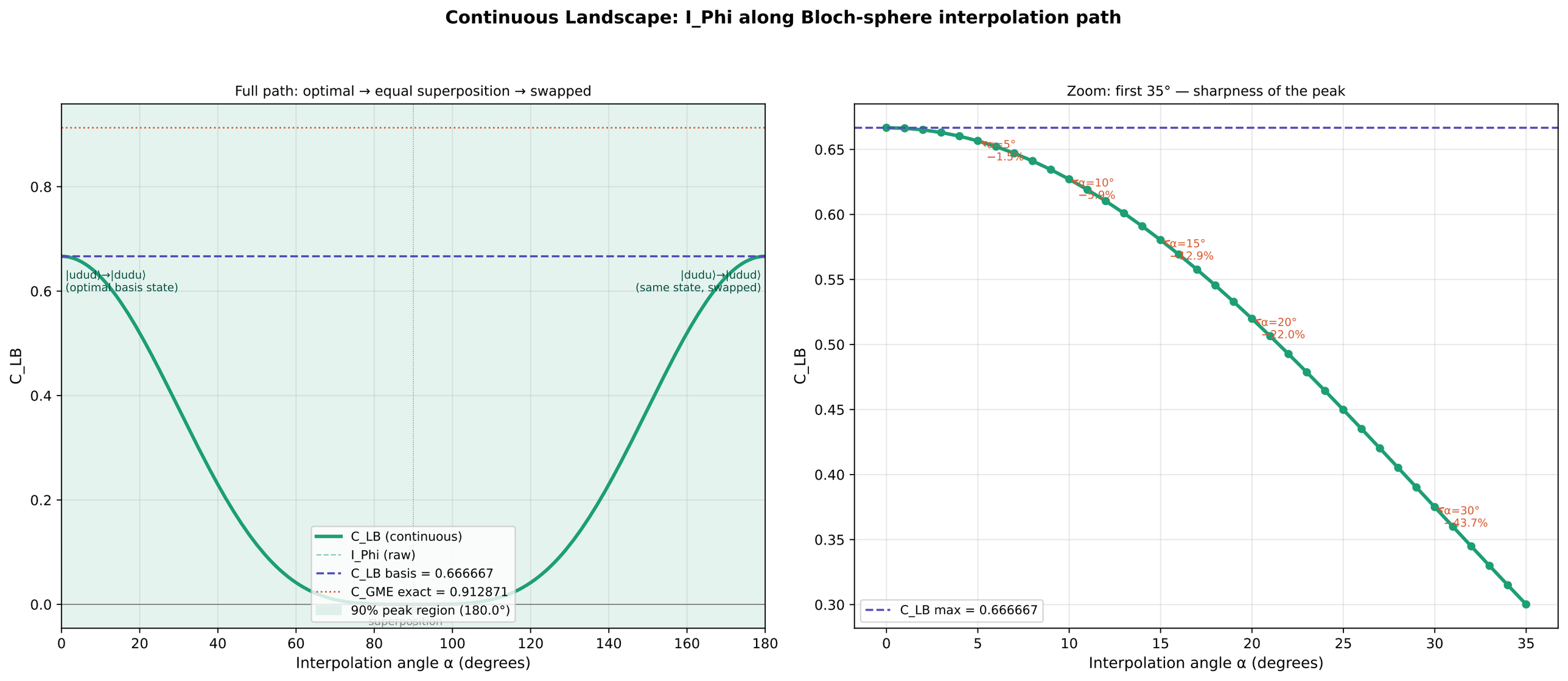
  z1 = |udud⟩
  z2 = |dudu⟩

  Pairs at maximum     : 2
  Physically unique    : 1
  (z1→z2 and z2→z1 are
   the same |Phi⟩, just
   labels swapped)

Continuous Landscape
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  Peak width at 90%    : 180.0°
  Peak character       : BROAD
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Continuous Landscape: I_Phi along Bloch-sphere interpolation path
Full path: optimal → equal superposition → swapped
|ududu⟩→|dudu⟩
(optimal basis state)
|dudu⟩→|ududu⟩
(same state, swapped)
C_LB (continuous)
I_Phi (raw)
C_LB basis = 0.666667
C_GME exact = 0.912871
90% peak region (180.0°)
Superposition
C_LB
Interpolation angle α (degrees)
Zoom: first 35° — sharpness of the peak
α=5°
−1.5%
α=10°
−5.9%
α=15°
−12.9%
α=20°
−22.0%
α=30°
−43.7%
C_LB max = 0.666667
C_LB
Interpolation angle α (degrees)

Summary: Answer to Supervisor's Question

Supervisor's question:
  'Is the maximum obtained only for the basis state, or do you also reach
   it for other states Phi? Are there single optimal points in the parameter
   space, or multiple optimal points?'

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1. DISCRETE (basis-state) SEARCH [240 pairs checked]

   2 pairs achieve the maximum I_phi = 0.66666667:
     |udud⟩ → |dudu⟩
     |dudu⟩ → |udud⟩

   These are physically IDENTICAL — z1 and z2 labels swapped.
   -> ONE unique optimal |Phi⟩ in basis-state space.

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2. CONTINUOUS LANDSCAPE (real states, Bloch-sphere interpolation)

   The peak is SHARP and ISOLATED at the basis state:

   alpha =   5°   C_LB = 0.656577   (+1.5% from max)
   alpha = 10°   C_LB = 0.627068   (+5.9% from max)
   alpha = 15°   C_LB = 0.580342   (+12.9% from max)
   alpha = 20°   C_LB = 0.519819   (+22.0% from max)
   alpha = 30°   C_LB = 0.375000   (+43.7% from max)
   alpha = 45°   C_LB = 0.166667   (+75.0% from max)
   alpha = 90°   C_LB = 0.000000   (+100.0% from max)

   Peak width at 90% of maximum: 180.0°
   -> BROAD peak

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CONCLUSION:

   SINGLE isolated optimal point in parameter space.
   The basis-state |udud⟩ ↔ |dudu⟩ is the unique global maximum.
   Moving even a few degrees away on the Bloch sphere reduces I_Phi.
   No other family of states achieves the same value.

   This confirms: the Hubbard ground state has Neel-type entanglement
   concentrated in the coherence between the two antiferromagnetic
   configurations. The basis-state approach is already exactly optimal.